

DIGITAL ELECTRONICS & LOGIC DESIGN PROJECT

Design & Simulation of 8-Bit Signed Binary Representation

A Comprehensive Analysis of Sign-Magnitude, One's Complement, and Two's Complement Methods in Hardware Architecture

01. Fundamentals

Introduction to Signed Numbers & Binary Representation Constraints

Unsigned vs. Signed Numbers

Unsigned 8-Bit Binary

All 8 bits are dedicated strictly to representing magnitude. There is no designation for negative values in unsigned representation.

$$0 \text{ to } 2^8 - 1 = [0, 255]$$

Use Case: Memory addressing, counter registers, pixel intensities.

Signed 8-Bit Binary

The Most Significant Bit (MSB, Bit 7) is designated as the **Sign Bit**. The remaining 7 bits dictate representation magnitude.

$$\text{MSB} = 0 \rightarrow \text{Positive (+)}, \text{MSB} = 1 \rightarrow \text{Negative (-)}$$

Use Case: ALU operations, digital signal processing (DSP), temperature metrics.

Sign-Magnitude Method

In Sign-Magnitude, Bit 7 indicates the sign while Bits [6:0] hold the raw binary magnitude.

Examples for Decimal ± 25 :

+25 → 0 0011001 (MSB = 0)

-25 → 1 0011001 (MSB = 1)

Key Limitation: Contains dual zero values ($+0 = 00000000$ and $-0 = 10000000$), requiring extra hardware logic for zero detection in ALUs.

 Digital logic circuit diagram simulation software technology banner

One's Complement Method



Bit Inversion

Positive values retain standard binary representation. Negative values are derived by flipping every bit ($0 \rightarrow 1$, $1 \rightarrow 0$) of the positive counterpart.



End-Around Carry

Addition requires an extra step: if addition produces a carry-out from the MSB, this carry bit must be added back to the LSB (End-Around Carry).



Double Zero Issue

Like Sign-Magnitude, One's Complement maintains two representations for zero (00000000 for +0 and 11111111 for -0).

02. Two's Complement

The Standard Architecture for Modern Computer Systems

8-Bit Two's Complement Capacity

256

Distinct Integer Values

Two's Complement Algorithm

Obtained by taking One's Complement (inverting all bits) and adding 1 to the LSB:

$$\text{Two's Complement} = \bar{A} + 1$$

Expressible Range: -128 to +127

Key Advantage: Unique representation for zero (00000000) and seamless hardware binary addition without end-around carry!

Representation Comparison

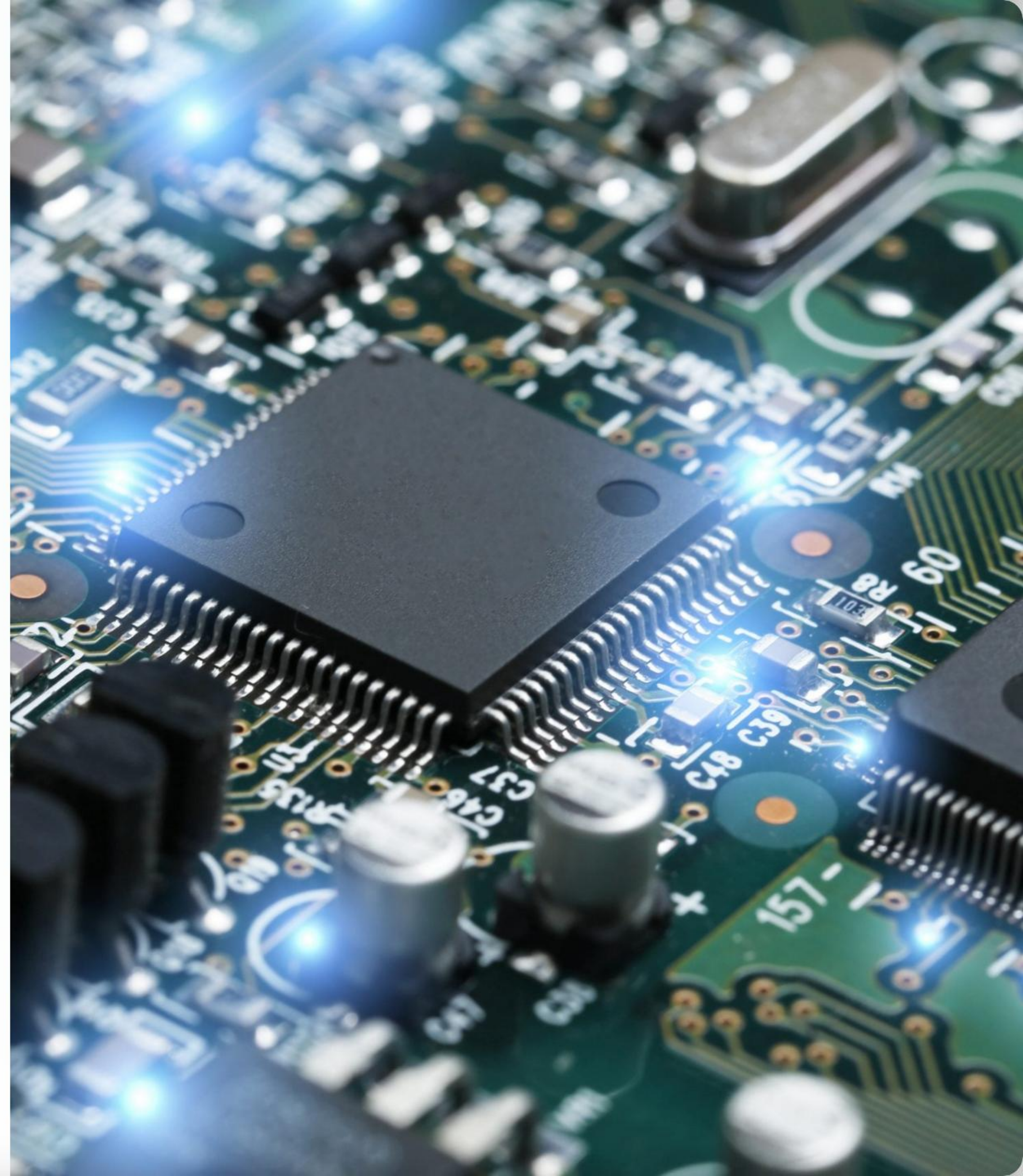
Decimal Value	Sign-Magnitude (8-Bit)	One's Complement (8-Bit)	Two's Complement (8-Bit)
+45	00101101	00101101	00101101
-45	10101101	11010010	11010011
+0	00000000	00000000	00000000
-0	10000000	11111111	00000000 (Unique!)
-128	N/A (Out of Range)	N/A (Out of Range)	10000000

Hardware Implementation

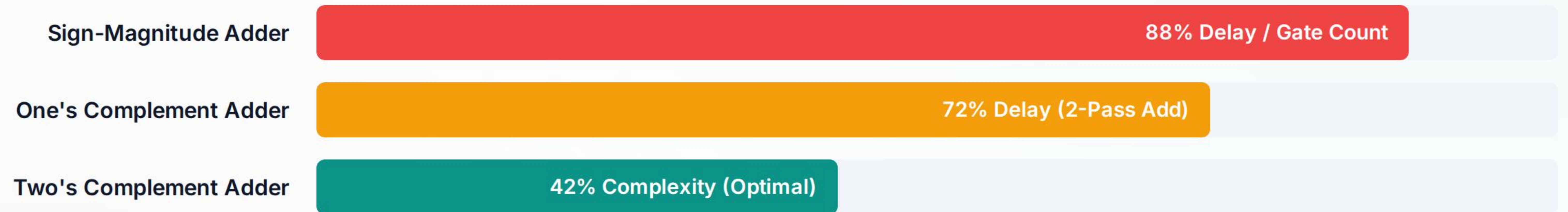
Adder / Subtractor Circuit Design

Two's Complement enables subtraction $A - B$ via addition $A + (-B)$ using XOR gates and standard Ripple Carry or Carry Lookahead Adders.

When Control Line $SUB = 1$, XOR gates invert input B, and carry-in $C_{in} = 1$ adds the required +1 automatically.



Hardware & Logic Complexity



Two's Complement requires significantly fewer logic gates and reduces propagation delay by unifying addition and subtraction into a single adder datapath.

Simulation & Verification Flow

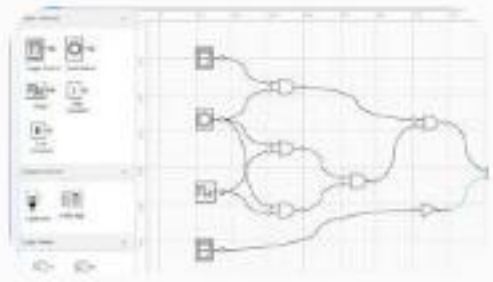


Questions & Discussion

Thank you for reviewing our Digital Design Project presentation.

🔗 Repository: `github.com/student-group/8bit-signed-binary-simulation`

Image Sources



<https://www.syncfusion.com/free-tools/wp-content/uploads/sites/22/2025/08/logic-circuit-simulator-v1.webp>

Source: www.syncfusion.com



<https://ntt.edu.vn/wp-content/uploads/2024/04/closeup-electronic-circuit-board-with-cpu-microchip-electronic-components-background-scaled.jpg>

Source: ntt.edu.vn